ILLOVO Share Price Analysis

Illovo Sugar Malawi Plc | Malawi Stock Exchange | 2 January 2024 to 29 October 2026

Summary of the analysis

This report explains the results from the R scripts in simple words. It uses the output tables and charts produced by the project. You do not need to know statistics to follow it.

- **The price went up a lot.** The closing price rose from MWK 1,260 to MWK 3,502. That is a rise of about 178%, so the price almost tripled in under three years.
- **The price moves like stairs, not like a smooth line.** It stays flat for weeks or months, then jumps. Since about late May 2026 the reported price has barely moved at all, staying near MWK 3,500.
- **The data is thin.** Only 63% of weekdays have a market report, and more than half of all price changes are exactly zero. This affects every result below.
- **Past prices tell us very little about the next price.** Today's change has almost no link to tomorrow's change. The tests agree that the price wanders, and only its changes are stable.
- **No forecasting model beat the simplest guess.** The guess "tomorrow's price will be the same as today's" had smaller average errors than every ARIMA-type model. On weekly data the more complex models did much worse.
- **Risk (volatility) does come in waves, but the GARCH model did not capture it well.** The chosen model still failed its own checks and predicted risk worse than a plain constant number.
- **The 20-day forecast should not be trusted as it stands.** It says MWK 3,508 rising to MWK 3,656, but the "likely range" around it is almost zero width. Real uncertainty is far larger.

Data source

The data comes from the Malawi Stock Exchange daily market reports. For each report day we have the closing price and the number of shares traded. All the data checks passed: no missing prices, no duplicate dates, no weekend dates and no negative volumes.

Item	Result	What it means
Period	2 Jan 2024 to 29 Oct 2026	Almost three years
Daily reports	466	Days with a usable report
Weekdays in the period	738	Days the market could have been open
Weekdays with no report	272	Missing from our data
Report coverage	63.1%	About 6 in 10 weekdays are covered
Returns that span a gap	171	The change covers more than one day
Zero returns	51.6%	Half of the price changes are zero
Reports with shares traded	77.7%	About 1 in 5 reports shows no trading
Very large moves (over 3 SD)	8	A few extreme jumps

What is a "return"? A return is the percentage change in price from one report to the next. If the price goes from 1,000 to 1,010, the return is 1%. The project uses log returns, which are almost the same as ordinary percentage changes for small moves.

Why coverage matters. A missing report does not always mean the market was closed. Because 37% of returns cover a gap, some "one-day" changes are really several days of change bundled together. Even so, in most months between 45% and 90% of weekdays are covered, so the gaps are spread across the whole period.

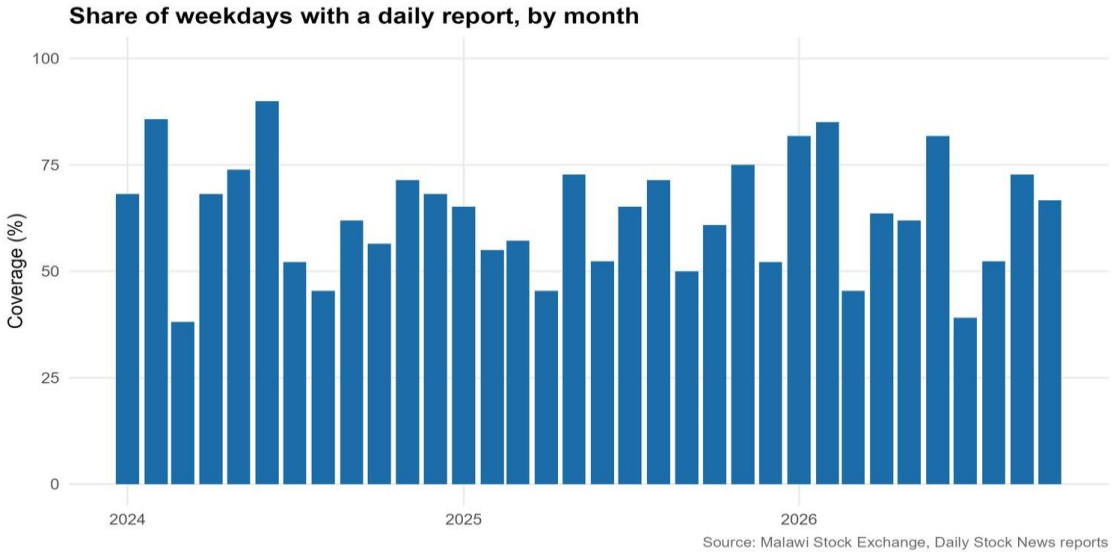


Figure 1: Share of weekdays with a report, by month

Share price statistics (2024 – 2026)

Year	Reports	First price (MWK)	Last price (MWK)	Change	Zero returns	Avg shares per report
2024	**169**	**1,260**	**1,355**	**+7.5%**	**67%**	**5,706**
2025	**157**	**1,355**	**2,330**	**+72.0%**	**45%**	**14,030**
2026 (to 29 Oct)	**140**	**2,330**	**3,502**	**+50.3%**	**41%**	**14,027**

As seen in the table:

- 2024 was quiet. The price barely moved and two thirds of the price changes were zero.
- 2025 was the big year, with a rise of 72%. Trading became more active too, with more than double the shares traded.
- 2026 added another 50% by the end of October, but most of that came before late May. After that the reported price is almost flat.

The chart below shows the staircase pattern. Each step is a sudden jump followed by a long flat stretch. The bottom panel shows that trading volume is mostly small, with a few very large days (the biggest was 931,048 shares). The typical report has only about 890 shares traded, while the average is 11,010, because a few huge days pull the average up.

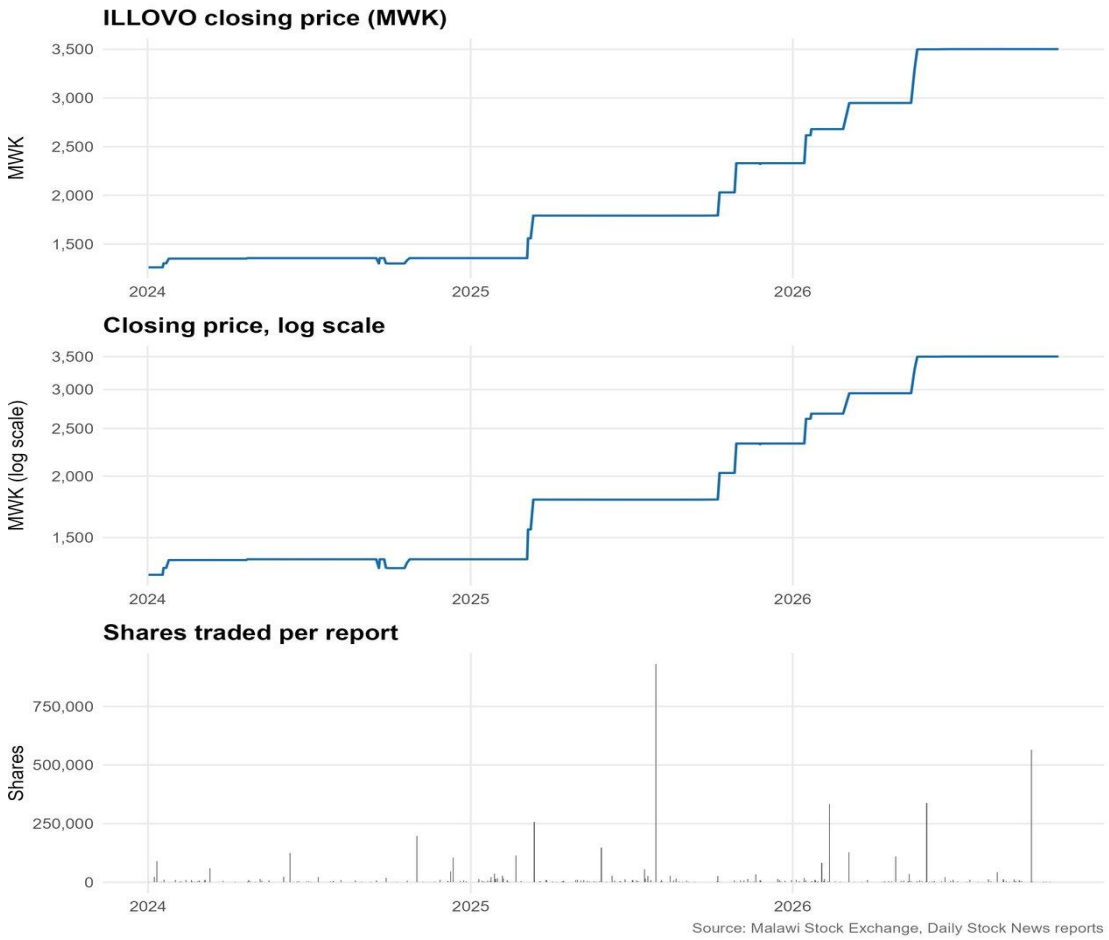


Figure 2: Closing price (top two panels) and shares traded per report (bottom)

How the returns behave

Measure	Daily-report returns	Weekly returns	Plain meaning
Average return	0.22%	0.70%	Average gain per report (or per week)
Typical swing (SD)	1.60%	2.92%	How much returns normally vary
Median	0%	0%	The middle value is "no change"
Biggest fall	-4.15%	-3.99%	Falls are small
Biggest rise	+13.98%	+17.10%	Rises are much bigger
Share of zero returns	51.6%	21.8%	Weekly data has fewer zeros

Falls are small, rises are big. The biggest fall is only 4.15% but the biggest rise is nearly 14%. The statistics confirm this: the skewness is +7.0 and the excess kurtosis is 52.6. In simple terms, most days are quite and a few days are extreme.

The Jarque-Bera test strongly rejects normal behaviour for both daily and weekly returns. In the chart below, the tall grey bar is the pile of zero returns and the red line is what a normal bell curve would look like.

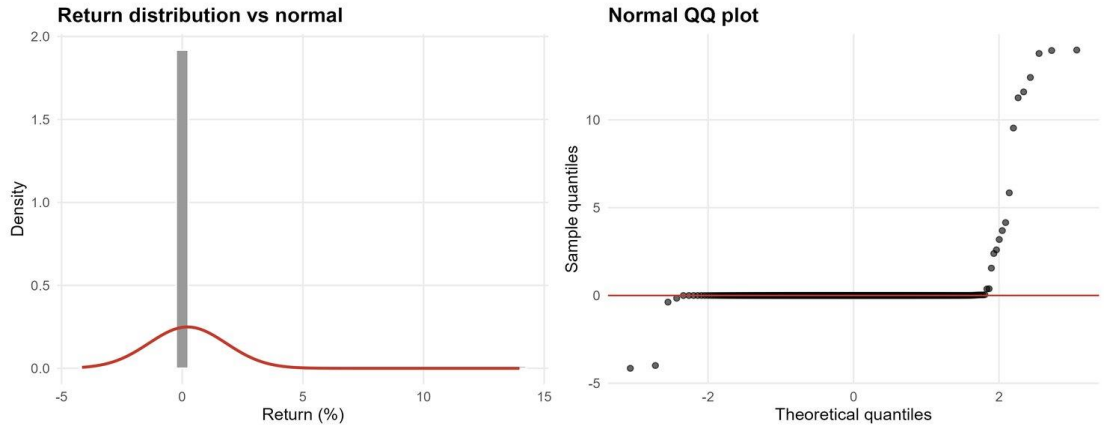


Figure 3: Returns compared with a normal curve (left) and a normality check (right)

Trading versus not trading

On reports with no shares traded, 90% of returns were zero. On reports with trading, only 40% were zero. So price changes mostly happen when shares actually change hands. Bigger trading volume also goes with bigger price moves (correlation 0.35), but when we look only at days with trading the link is weak (0.13).

The biggest moves

Date	Close (MWK)	Return	Shares traded	After a reporting gap?
13 Mar 2025	1,791.70	+13.98%	0	Yes (3 weeks)
7 Mar 2025	1,558.00	+13.95%	203	No
29 Oct 2025	2,330.00	+13.78%	1,599	Yes (2 weeks)
10 Oct 2025	2,030.00	+12.41%	2,000	Yes (2 weeks)
16 Jan 2026	2,616.60	+11.59%	7,139	Yes (2 weeks)
19 May 2026	3,300.00	+11.26%	54	Yes (2 weeks)
6 Mar 2026	2,948.35	+9.53%	128,466	Yes (5 weeks)

22 May 2026	3,498.62	+5.84%	3,627	Yes (3 weeks)

Seven of the eight biggest moves came right after a reporting gap. This suggests that many "big daily jumps" are really several days or weeks of price change shown at once. The 13 March 2025 jump of nearly 14% happened with zero shares traded.

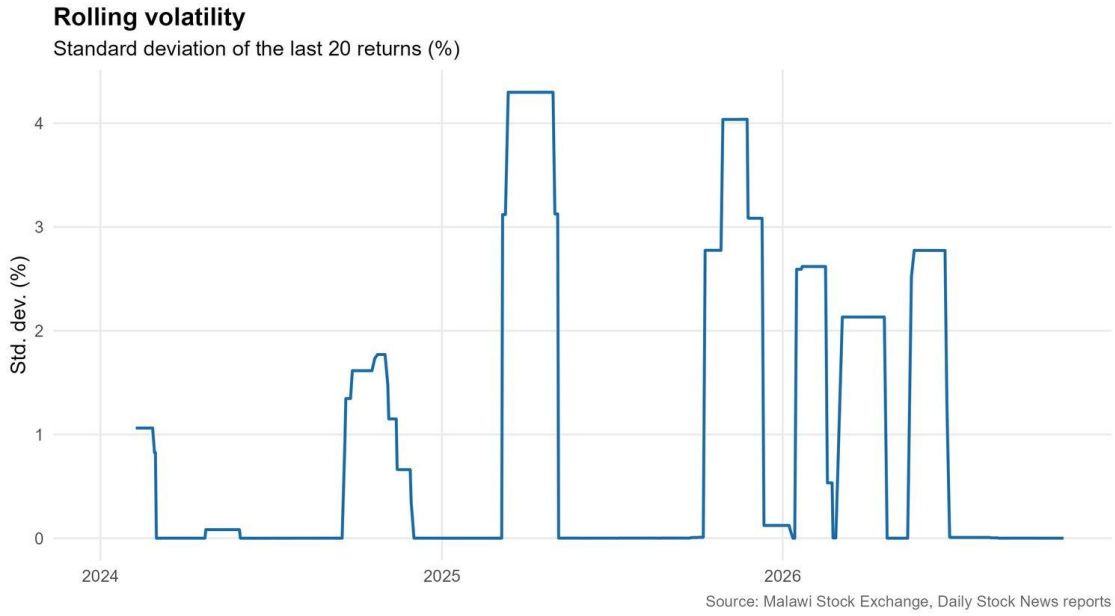


Figure 4: Rolling volatility (how much returns have been swinging over time)

Stationarity tests

A series is "stationary" if it keeps returning to a steady average level and behaves the same way over time. A price that keeps climbing is not stationary. Most forecasting models only work properly on stationary data, so we test first.

Three tests were used. Each gives a p-value. A small p-value (below 0.05) is strong evidence against the starting assumption of the test. Note that the ADF and PP tests start by assuming the series is NOT stable, while the KPSS test starts by assuming it IS stable. When the tests point the same way, we can be confident.

Series	ADF	Phillips-Perron	KPSS	Conclusion
Daily price (log)	0.496	0.743	0.010	Not stable
Daily returns	0.010	0.010	0.100	Stable
Weekly price (log)	0.468	0.740	0.010	Not stable
Weekly returns	0.010	0.010	0.100	Stable

Result: All three tests agree. The price level wanders and does not settle, but the returns (the changes) are stable. This is why the models use returns and not prices. The KPSS results also say the price needs to be differenced once, which means we should work with changes in price. This is the usual finding for share prices.

Patterns in the returns

Autocorrelation checks

The daily Ljung-Box test gives p = 0.022 at 10 lags, a small sign of a pattern, but at 20 lags the p-value is 0.31. The correlation between one return and the next is only 0.026, which is close to zero. If we look only at days when the price actually moved, it is 0.19, a mild tendency for moves to follow moves in the same direction. For weekly returns there is no pattern at all (p = 0.71).

Volatility clustering

The test on squared returns (p = 0.001) and the ARCH test (p = 0.0015 at 5 lags and 0.0002 at 10 lags) show that big moves tend to bunch together. This is common in share markets and is the reason we try GARCH models. For weekly data there is no clear clustering (p = 0.85 and 0.52).

Thin trading

After a zero return, the next return is also zero 58% of the time. After a non-zero return, the next one is zero only 45% of the time. The test (p = 0.006) says this is not luck. The price tends to "sleep" for a while, then wake up. This is typical of a market where trading is not frequent.

Structural breaks

- **Price level:** three break dates were found: 10 March 2025, 27 October 2025 and 15 May 2026. These match the big upward jumps in the staircase chart.
- **Volatility:** no break was found. The size of the swings stayed fairly similar.
- **Average return (CUSUM test):** p = 0.23, so there is no evidence that the average return became unstable.

Forecasting with ARIMA-type models

To test forecasting skill honestly, the last 20% of the data was hidden (93 daily reports and 30 weekly observations). The models were built on the first 80%, then asked to predict one step ahead, again and again. The predictions were compared with what really happened.

The error measures are RMSE and MAE. Smaller is better. They are shown as decimals, so 0.013 means 1.3%. The last column shows the average error in Malawi kwacha when the return forecast is turned back into a price.

Daily results

Model	RMSE	MAE	Hit rate	Avg price error (MWK)
No change (zero return)	0.01315	0.00185	n/a	5.96
Average past return (drift)	0.01303	0.00402	75.9%	13.29
AR(1)	0.01319	0.00410	74.1%	13.58
MA(1)	0.01315	0.00408	74.1%	13.50
AR(2) = auto ARIMA(2,0,0)	0.01339	0.00402	74.1%	13.36
MA(2)	0.01336	0.00404	70.4%	13.42
ARMA(1,1)	0.01348	0.00423	74.1%	14.01

All models have almost the same RMSE (about 1.3%), because that is dominated by the few huge jumps that no model predicts. But on the average error (MAE) the "no change" guess is more than twice as good (0.00185 against about 0.004). In kwacha, it is wrong by about MWK 6 on average, while the other models are wrong by about MWK 13 to 14. The reason is simple: more than half the time the price does not change, and the models keep predicting a small rise.

A hit rate of 74% to 76% looks impressive, but be careful. The models almost always predict a small rise, and over this period many returns were zero or positive. A high hit rate here is not clear proof of skill.

The Diebold-Mariano test compares the best automatic model with the no-change guess. For daily data p = 0.36, which means there is no real difference between them.

Weekly results

Model	RMSE	MAE	Hit rate	Avg price error (MWK)
No change (zero return)	0.03122	0.00574	n/a	18.46
Average past return (drift)	0.03073	0.01245	73.1%	41.38
AR(1)	0.03138	0.01224	73.1%	40.82
ARMA(1,1)	0.04807	0.04001	46.2%	138.49
Auto ARIMA(3,0,2)	0.07604	0.06682	46.2%	230.81

The automatic ARIMA(3,0,2) model made errors more than twice as large as the simple ones, and its price error was about MWK 231 against MWK 18 for "no change". The Diebold-Mariano test confirms that this is a real difference (p = 0.0001), and it works against the complex model.

On the training data, the weekly ARMA(1,1) looked best (lowest AICc and BIC). But on new data it did badly. This is called overfitting: the model learned the quirks of the old data instead of a real pattern.

After fitting, the daily model residuals show no leftover autocorrelation (p = 0.12), but the squared residuals still show clustering (p = 0.001). This is the signal that a model of changing risk (GARCH) may help.

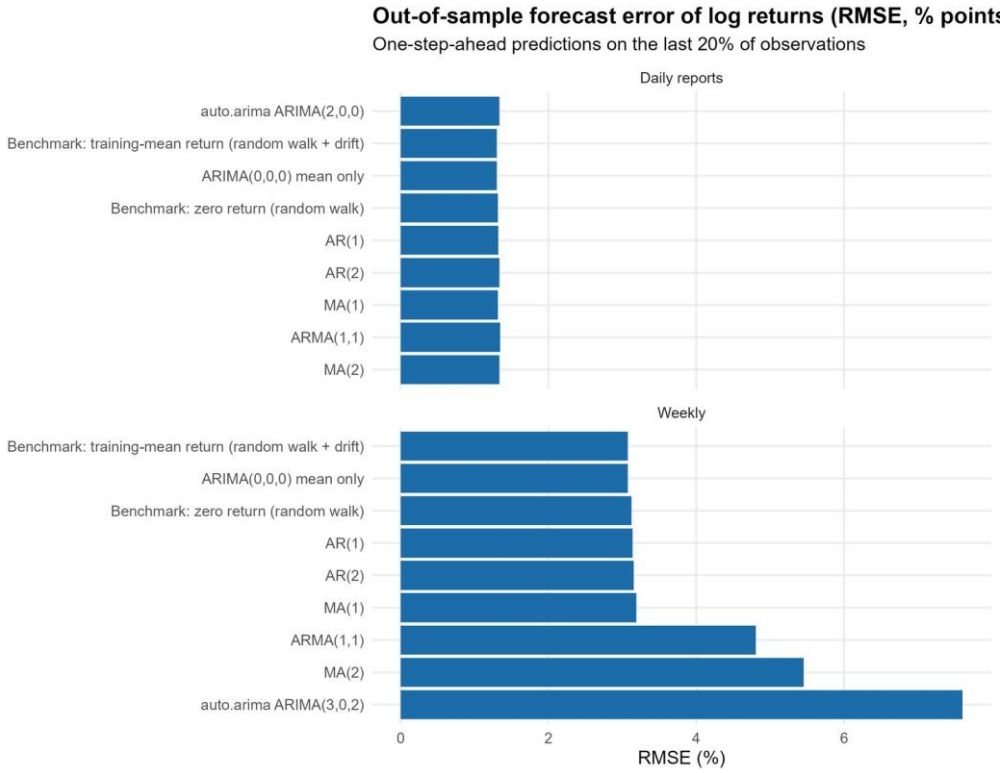


Figure 5: Out-of-sample forecast errors by model

Modelling changing risk with GARCH

GARCH models try to describe how risk (volatility) rises and falls over time. Five versions were compared. Lower AIC and BIC values mean a better fit.

Model	AIC	BIC	Comment
GARCH(1,1), normal	-219	-204	Poor fit
GARCH(1,1), Student-t	-3,392	-3,373	Much better
GARCH(1,1), skewed Student-t	-3,434	-3,411	Better again
GJR-GARCH(1,1), Student-t	-3,450	-3,427	Lowest (chosen)
EGARCH(1,1), Student-t	-1,518	-1,494	Worse

Models that allow for extreme moves (Student-t) fit far better than the normal model. This confirms that big jumps are a normal part of this share.

But there are serious problems with the chosen GJR-GARCH model:

- **It still fails its checks.** The test on residuals gives p = 0.0099, the test on squared residuals gives p = 0.0010 and the ARCH test gives p = 0.0015. All are below 0.05, so the model has left patterns behind.

- **It found almost no risk "memory".** The main GARCH parameters (alpha and beta) are practically zero. This means the model is not tracking risk through time. It is acting like a constant-risk model with very heavy tails.
- **Extreme tail setting.** The shape parameter is about 2.1, close to 2. That is the edge where the maths says risk becomes unbounded. It is a sign the model is struggling with the data.
- **Unstable key estimate.** The "leverage" term (gamma) is 0.97 but its standard error is about 42 (p = 0.98), so it cannot be trusted.
- **It predicts risk worse than a constant.** In the test period, the plain constant-variance guess scored far better (QLIKE 1.6) than every GARCH model (from about 1,400 up to hundreds of millions). Lower is better.

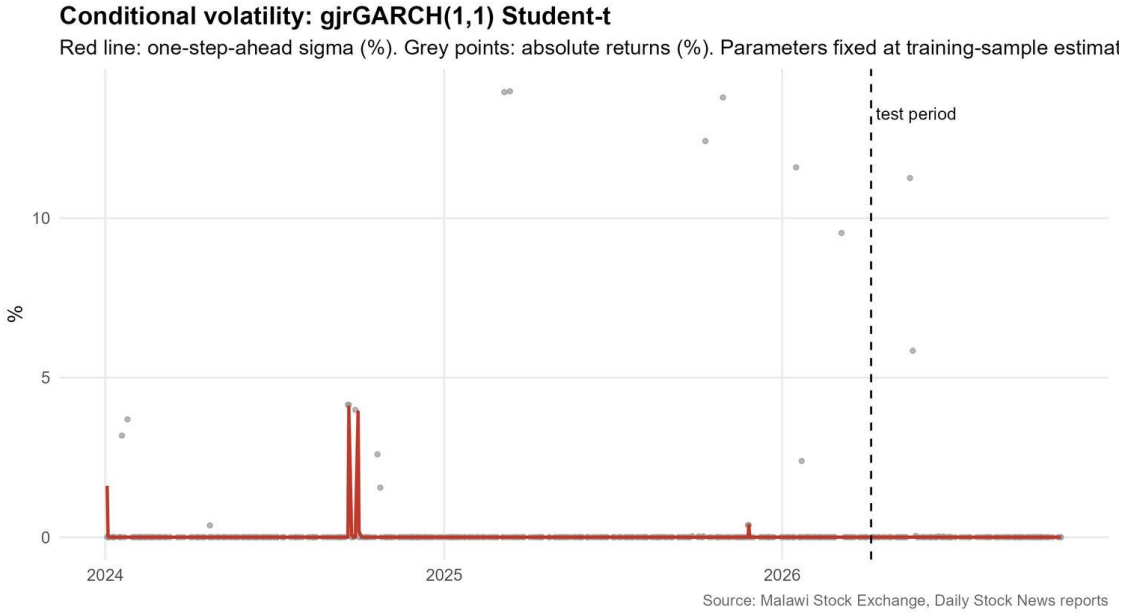


Figure 6: The fitted risk line (red) sits near zero except for a few spikes, while real moves (grey dots) are large

The chart shows the problem clearly. The red line, which is the model's estimate of risk, stays almost flat at zero. The grey dots, which are the actual absolute returns, are often far above it. With about half the returns equal to zero and many gaps, GARCH models struggle, because they assume a regular flow of daily data.

The 20-step price forecast

Item	Result
Last reported price (29 Oct 2026)	MWK 3,502
First forecast (30 Oct 2026)	MWK 3,508

20th forecast (26 Nov 2026)	MWK 3,656
Total rise over 20 report days	about 4.4%
95% range at step 1	MWK 3,508.16 to 3,508.38
95% range at step 20	MWK 3,655.49 to 3,656.86

Where does the rise come from? The forecast adds about 0.22% each report day, which is simply the average return in the past data. That average was pulled up by a few big jumps. So the forecast is basically "the past average growth will continue".

The ranges are far too narrow. A proper 95% range should be a zone that the price would stay inside 95 times out of 100. Here the range at step 20 is only about MWK 1.4 wide, which is a margin of roughly 0.02%. Yet the share's normal swing is about 1.6% per report day, which is around MWK 56 at today's price. The model's daily risk estimate is about 0.002%, hundreds of times too small. This comes from the GARCH problems explained above.

A rough, more realistic picture. If we simply use the normal 1.6% swing, 20 report days could move the price by roughly 7% either way (1.6% x the square root of 20). A 95% range would then be about plus or minus 14%, or roughly MWK 3,000 to MWK 4,200. This is only a back-of-the-envelope illustration, not a model result, but it shows how much wider the true uncertainty is. It also does not account for the fact that the price often does not move at all and then jumps.

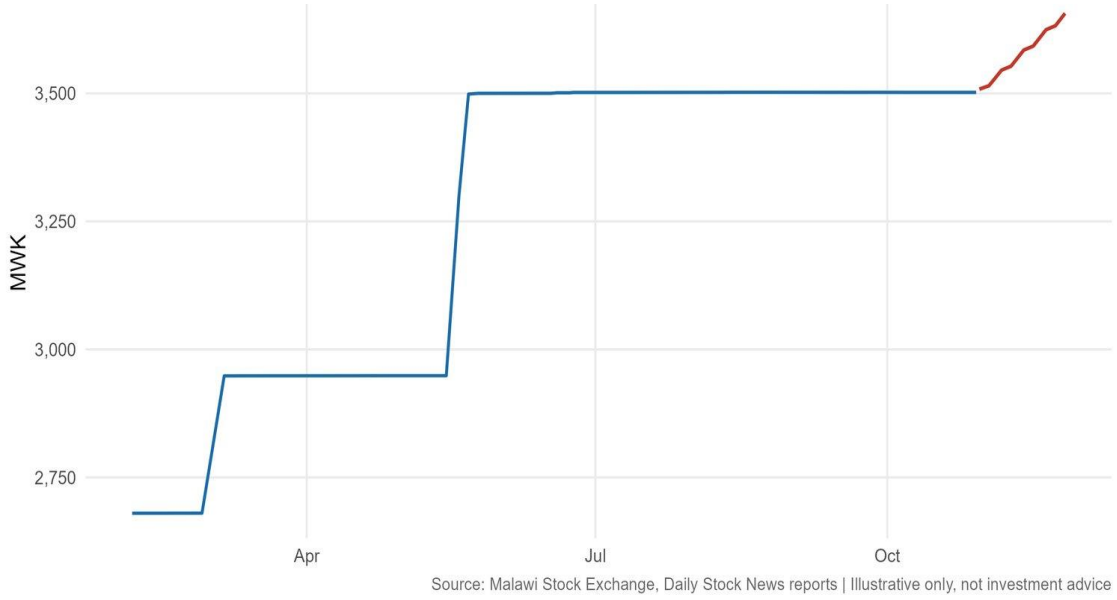


Figure 7: Last 120 reports and the 20-step forecast. The forecast bands are too thin to see.

Limits

Main limits

- Missing reports: 37% of weekdays have no report, so the "daily" data is really uneven in time.
- Thin trading: half of the returns are zero and the price moves in jumps. Most standard models assume smooth, frequent changes.
- Extreme values: a handful of jumps drive most of the statistics (average, spread, kurtosis).
- Outside factors: dividends, company news, sugar prices, inflation and the exchange rate are not in the models.
- Short history: under three years of data and a small test set, especially for weekly data (30 points).

Suggested next steps

- Check the largest moves and the 10 price changes with zero volume against the original MSE PDFs.
- Fix the forecast bands. Use a simple method based on past returns, such as resampling actual returns, or use the real sample volatility, so that the range is believable.
- Keep "no change" as the benchmark and show it next to every model. Beating it is the real test.
- Try weekly or monthly data only, which avoids many gaps and zeros, and add a rolling-window test.
- Add dividend dates and see whether the jumps line up with announcements or results.
- Consider models made for thin trading (for example, models that treat zero returns separately from non-zero returns).

Plain-English glossary

Term	Meaning
Return	The percentage change in price from one report to the next.
Volatility	How much the price swings. High volatility means large ups and downs.

Stationary	Behaves in a steady way over time. Prices usually are not, returns usually are.
p-value	A number between 0 and 1. Below 0.05 usually means the pattern is unlikely to be pure luck.
Autocorrelation	Whether one day's change is linked to the next day's change.
Volatility clustering	Calm periods and wild periods come in groups.
Heavy tails	Extreme moves happen more often than a bell curve would suggest.
Thin trading	Shares trade rarely, so many days show no price change.
Random walk	The next price is today's price plus an unpredictable change.
ARIMA / AR / MA	Models that predict the next value from past values or past errors.
GARCH	A model for how risk rises and falls over time.
RMSE / MAE	Two measures of average forecast error. Smaller is better.
Overfitting	A model that fits old data very well but fails on new data.
Out-of-sample	Tested on data the model has not seen when it was built.

Source data: Malawi Stock Exchange, Market Reports (https://mse.co.mw/market/reports).
All interpretations are based on the project output tables and charts.